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Review

Geometric deep learning for drug discovery

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ABSTRACT

Drug discovery is a time-consuming and expensive process. With the development of Artificial Intelligence (AI) techniques, molecular Geometric Deep Learning (GDL) has recently emerged for learning from molecules and accelerating drug discovery. However, there is a gap between researchers in the field of deep learning and biomedicine. This review provides a comprehensive overview of the recent literature on GDL for drug discovery based on molecular three-dimensional (3D) representation learning and symmetry learning, highlighting its applications of molecular property prediction, intermolecular interaction, molecular design, molecular conformation prediction, and molecular 3D pretraining. We discuss the challenges of current molecular 3D representation learning and tasks. Further, we propose future directions to promote drug discovery and deal with these challenges. The latest advances in GDL for drug discovery are summarized in a GitHub repository https://github.com/3146830058/Geometry-Deep-Learning-for-Drug-Discovery.

1. Introduction

Deep learning(LeCun et al., 2015; Schmidhuber, 2014) has made a significant impact on drug discovery, revolutionizing various stages of the process, and accelerating the identification and development of new therapeutic compounds. By leveraging the power of deep learning techniques, researchers can analyze large volumes of data, extract meaningful patterns, and make predictions to guide the drug discovery process more efficiently(Gupta et al., 2021; Pandey et al., 2022; Stokes et al., 2020; Zeng et al., 2022; Chakraborty and Hasija, 2023).

Previous molecular modeling methods are mostly driven by molecular representation of either 1D sequences(e.g. Simplified Molecular Line Entry System (SMILES), amino acid sequences) or 2D graphs (Öztürk et al., 2020). Nevertheless, molecular structures are known to be determinants of their functions, and a comprehensive understanding of molecular interactions necessitates the consideration of their 3D structures. GDL, deep learning methods incorporating geometric priors (i.e. information about the structure and symmetry properties of the input variables)(Atz et al., 2021; Bronstein et al., 2021), has made remarkable progress in modeling molecular structures(Jin et al., 2022; Jumper et al., 2021; Satorras et al., 2021; Somnath et al., 2022). Despite significant advances, utilizing graph deep learning (GDL) for drug discovery remains a complex endeavor. Two key factors drive this complexity. First, applying GDL requires extensive bioinformatics expertise. Second, researchers must navigate a diverse landscape of GDL algorithms. Consequently, a comprehensive survey reviewing the application of GDL for drug discovery would prove valuable by clarifying this burgeoning field. Such a survey is needed to guide researchers seeking to harness GDL to advance drug discovery efforts.

Overall, our contributions can be summarized as follows: (1) We summarize the recent application of GDL for drug discovery based on molecular 3D representation learning and symmetry learning (Fig. 1). (2) We introduce a new taxonomy to categorize the various tasks of GDL in drug discovery, elaborating on what these tasks are and how GDL solves them. (3) We propose future directions to promote drug discovery and deal with the challenges of current molecular 3D representation learning and tasks. (4) We create a public GitHub repository for collecting the papers and related open-source codes, keeping abreast of the latest research developments.

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2. Backgrounds

This section will cover molecular 3D representation learning and symmetry learning as a priori in understanding the application of GDL for drug discovery, based on which we summarized the surveyed works in Table 1, listing their 3D descriptors, generative models (if have), applications, and symmetry groups.

2.1. Molecular 3D representation learning

Representation learning refers to the process of learning representations or features of data that can aid in the extraction of relevant information for building predictors or classifiers(Bengio et al., 2012). Molecular data can be mainly represented as three descriptors in 3D spaces, i.e., 3D graphs, 3D surfaces, and 3D voxels (Fig. 2**a**).

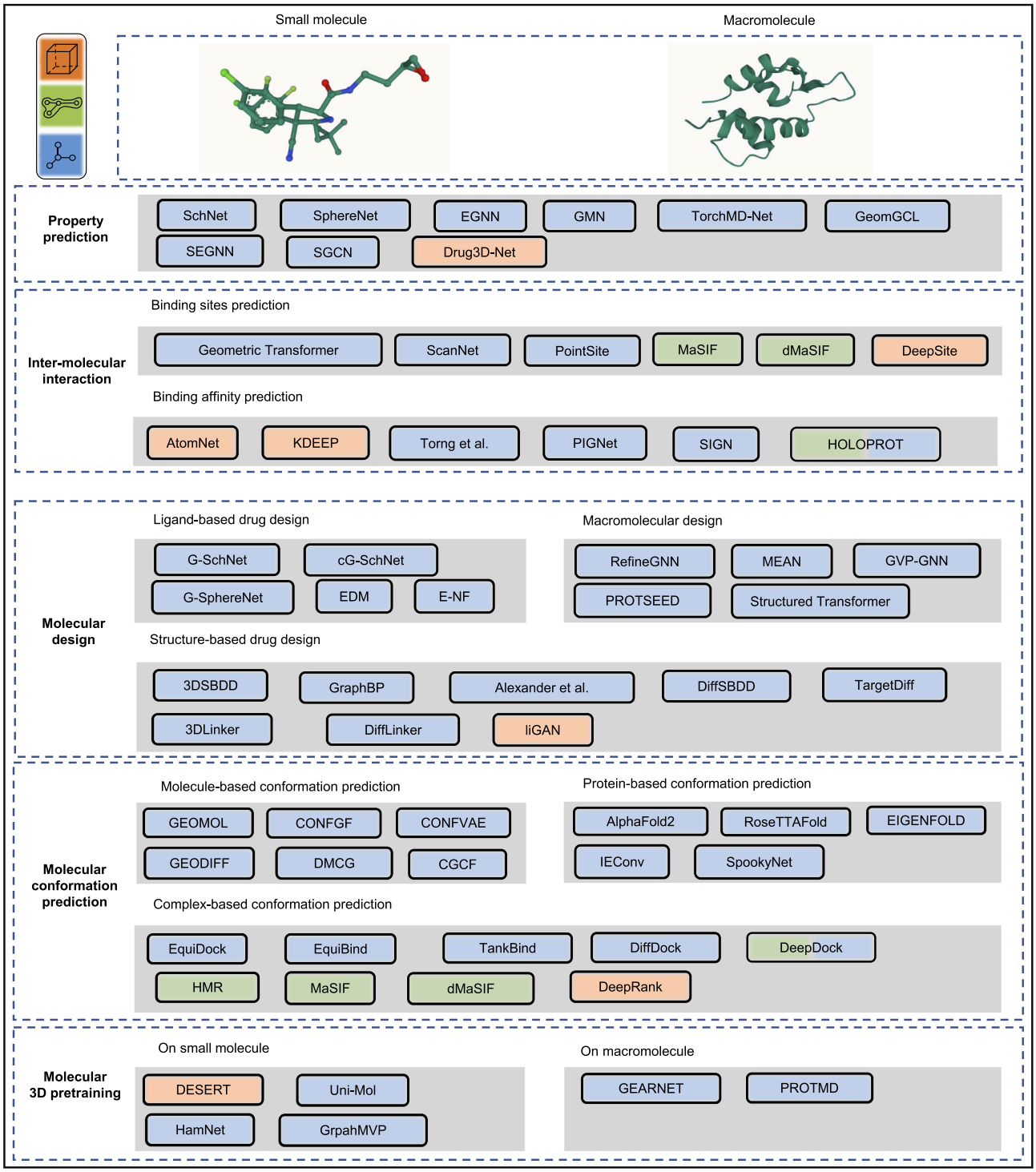

Fig. 1. Overview of GDL method on drug discovery in this review. We use different colors to represent different molecular descriptors used in literature(See top left corner. Orange for voxel-based methods, green for mesh-based methods, and blue for 3D-graph-based methods. Some methods use multiple descriptors and are therefore filled with multiple colors). We divide the tasks on drug discovery into five categories, including property prediction, intermolecular interactions, molecular design, molecular conformation prediction, and molecular 3D pretraining.

Table 1
Summary of representative molecular GDL methods.

Method	3D descriptor	Generative model	Application	Symmetry group
SchNet(Schütt et al., 2017)	3D graph	–	Molecular property prediction	E(3)
DimeNet(Gasteiger et al., 2020)	3D graph	–	Molecular property prediction	E(3)
EGNN(Satorras et al., 2021)	3D graph	–	Molecular property prediction	E(n)
Drug3D-Net(Li et al., 2021a)	3D voxels	–	Molecular property prediction	–
ScanNet(Tubiana et al., 2021)	3D graph	–	Binding site prediction	E(3)
DeepSite(Jiménez et al., 2017)	3D voxels	–	Binding site prediction	–
K_{DEEP}(Jiménez et al., 2018)	3D voxels	–	Binding affinity prediction	–
HOLOPROT(Somnath et al., 2022)	3D graph & 3D mesh	–	Binding affinity prediction	–
G-SchNet(Gebauer et al., 2019)	3D graph	Autoregressive model	LBDD	SE(3)
MolGym(Simm et al., 2020)	3D graph	Reinforcement learning	LBDD	SE(3)
cG-SchNet(Gebauer et al., 2021)	3D graph	Autoregressive model	LBDD	SE(3)
E-NF(Garcia Satorras et al., 2021)	3D graph	Flow-based model	LBDD	E(3)
G-SphereNet(Luo and Ji, 2022)	3D graph	Autoregressive model	LBDD	SE(3)
EDM(Hoogeboom et al., 2022)	3D graph	Diffusion model	LBDD	E(3)
GraphBP(Liu et al., 2022)	3D graph	Autoregressive model	SBDD	SE(3)
LiGAN(Ragoza et al., 2021)	3D voxels	VAE	SBDD	–
3DSBDD(Luo, 2022)	3D graph	Autoregressive model	SBDD	SE(3)
TargetDiff(Guan et al., 2023)	3D graph	Diffusion model	SBDD	SE(3)
Ig-VAE(Eguchi et al., 2020)	3D graph	VAE	Macromolecular design	SE(3)
Structured Transformer(Ingraham et al., 2019)	3D graph	Autoregressive model	Macromolecular design	SE(3)
GVP-GNN(Jing et al., 2021)	3D graph	Autoregressive model	Macromolecular design	SO(3)
RefineGNN(Jin et al., 2022)	3D graph	Autoregressive model	Macromolecular design	SE(3)
ProtDIFF(Trippe et al., 2023)	3D graph	Diffusion model	Macromolecular design	E(3)
MEAN(Kong et al., 2023)	3D graph	–	Macromolecular design	E(3)
PROTSEED(Shi et al., 2023)	3D graph	–	Macromolecular design	SE(3)
GraphDG(Simm and Hernández-Lobato, 2020)	3D graph	VAE	Molecule-based conformation prediction	SE(3)
ConfVAE(Xu et al., 2021)	3D graph	VAE	Molecule-based conformation prediction	SE(3)
ConfGF(Shi et al., 2021)	3D graph	Score matching	Molecule-based conformation prediction	SE(3)
DGSM(Luo et al., 2021)	3D graph	Score matching	Molecule-based conformation prediction	SE(3)
CGCF(Xu et al., 2021)	3D graph	Flow-based model	Molecule-based conformation prediction	SE(3)
GeoMol(Ganea et al., 2021)	3D graph	Optimal transport	Molecule-based conformation prediction	SE(3)
GeoDiff(Xu et al., 2022)	3D graph	Diffusion model	Molecule-based conformation prediction	SE(3)
MaSIF(Gainza et al., 2019)	3D mesh	–	Complex-based conformation prediction	SE(3)
dMaSIF(Sverrisson et al., 2021)	3D mesh	–	Complex-based conformation prediction	SE(3)
DeepDock(Méndez-Lucio et al., 2021)	3D mesh	–	Complex-based conformation prediction	SE(3)
DeepBSP(Bao et al., 2021)	3D voxels	–	Complex-based conformation prediction	–
EquiBind(Stärk et al., 2022)	3D graph	Optimal transport	Complex-based conformation prediction	SE(3)
DiffDock(Corso et al., 2023)	3D graph	Diffusion model	Complex-based conformation prediction	–
HMR(Wang et al., 2023)	3D mesh	–	Complex-based conformation prediction	SE(3)
AlphaFold2(Jumper et al., 2021)	3D graph	–	Protein-based conformation prediction	SE(3)
EIGENFOLD(Jing et al., 2023)	3D graph	Diffusion model	Protein-based conformation prediction	SE(3)

2.1.1. 3D graphs

3D graphs are mathematical objects that consist of three components: a set of nodes denoted by $\mathscr{V}$, a set of edges denoted by $\mathscr{E}$, and a set of information about node positions in 3D space denoted by $\mathscr{R}$. The nodes can represent atoms, residues, or other molecular entities, and the edges can correspond to chemical bonds, distances between atoms, or other criteria such as nearest neighbors. These components can be combined in various ways to represent different molecular structures and relationships. Graph descriptors are commonly processed using graph neural networks (GNNs)(Zhou et al., 2020). GNNs employ a message passing (MP) process that iteratively updates node features to obtain a more informative representation(Gilmer et al., 2017). In the case of 3D graphs, node features encompass 3D positional information, which can be referred to as the geometric message passing process (Fig. 2**b**). This process can be mathematically expressed as:

$$m_{ij} = \psi_m\left(h_i^l, h_j^l, a_{ij}\right)$$
$$h_i^{l+1} = \psi_h\left(Aggr\left(m_{ij_{j\in\mathscr{N}(i)}}\right), h_i^l\right)$$

where h_i^l is the feature of i-th node of the l-th network layer, m_{ij} is the output by the message passing on the edge (i,j), a_{ij} is conditioned on geometric information, ψ_m and ψ_h represent learnable functions, and $Aggr$ is a permutation-invariant aggregation operation (e.g., mean or sum).

2.1.2. 3D surfaces

3D surfaces of molecules are described as a set of smooth distance functions. In computer graphics and vision applications, surfaces are commonly represented as triangulated meshes. Such meshes are topological structures consisting of a collection of triangles, which can be viewed as a piecewise planar approximation of a surface constructed by joining triangles along their edges(Bronstein et al., 2021). The mesh vertices are endowed with geometric and chemical attributes that encode important properties (Fig. 2**c**). Geodesic convolutions and GNNs are popular techniques for extracting features from these descriptors.

2.1.3. 3D voxels

3D voxels are a type of geometric representation that discretizes 3D space into regular cells, each containing some information about the underlying object or scene. Each channel of 3D voxels can represent distinct properties (e.g. hydrophobic, hydrogen-bond donor or acceptor, aromatic, etc.)(Jiménez et al., 2018; Ragoza et al., 2021). We can also voxelize molecular shapes using one single channel(Long et al., 2022). The 3D Convolutional Neural Networks (3D CNNs) serve as an extension to the conventional 2D CNNs (Fig. 2**d**). Initially, 3D filters are employed for convolution operations to capture local patterns within the data. Subsequently, pooling is utilized to reduce the spatial dimensions, decrease the computational complexity, and enhance the model's generalization capabilities. Lastly, the fully connected layer integrates the high-level features extracted by the convolutional and pooling layers for decision-making tasks, such as classification and regression.

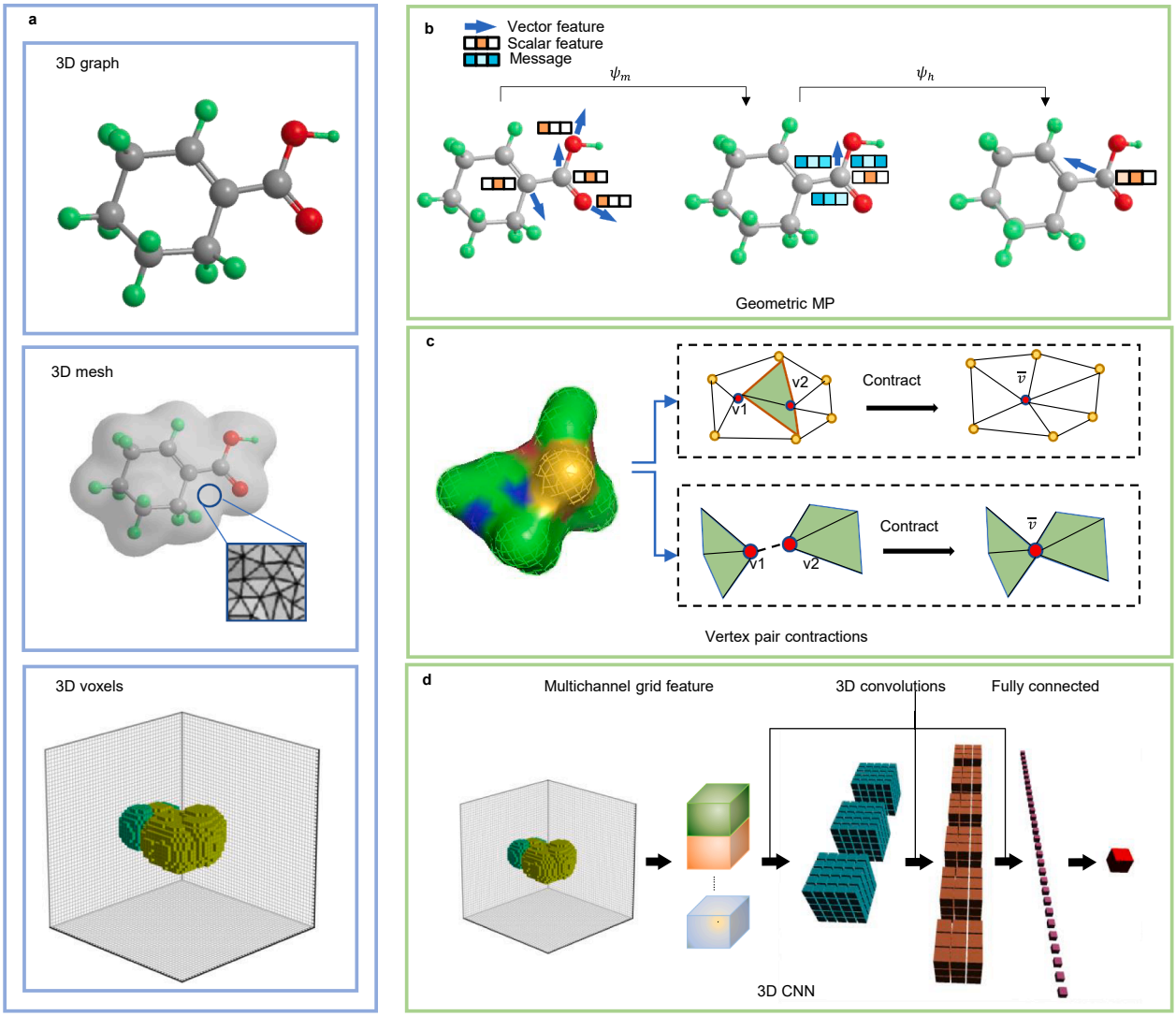

Fig. 2. Molecular descriptors and representation learning. (a). Molecular descriptors of 3D graph, 3D mesh, and 3D voxels, illustrated for 1-Cyclohexenecarboxylic Acid. **(b).** Geometric message passing process for molecular descriptors of 3D graph and 3D mesh. ψ_m represents the stage of message computation and ψ_h represents the stage of aggregation and update. Both the scalar feature and the vector feature (optional) are involved in message passing process. **(c).** Contractions of vertex pairs. The contraction of vertex pairs, which encompasses edge contraction as well as non-edge contraction, is employed to systematically decrease the number of faces and vertices inherent in the original mesh. This process is instrumental in the optimization and simplification of complex geometric structures, contributing to more manageable computational models and streamlined analyses(Li et al., 2022). **(d).** General 3D CNN frame for the molecular descriptor of 3D voxels. Given the input of multiple channels representing multiple properties, 3D convolution operation, and pooling operation are used to extract features from those voxels and reduce their size, fully connected layer integrates the features for target tasks.

2.2. Symmetry learning

Given a function $\mathscr{F} : \mathscr{X} \rightarrow \mathscr{Y}$ and a transformation $\mathscr{T}$ (eg. rotation, translation, reflection or permutation) of an acting symmetry group $\mathscr{G}$, symmetry generally refers to invariance or equivariance, which are used to describe the behavior $\mathscr{F}$ with respect to $\mathscr{G}$(Atz et al., 2021). Some of the most common symmetry groups are Euclidean group E(3), special Euclidean group SE(3), and permutation group. Both E(3) and SE(3) consist of the transformation of rotation and translation in 3D space, and E(3) additionally contains the transformation of reflection (Fig. 3). The formulations of invariance and equivariance are introduced below:

Equivariance. A function $\mathscr{F}$ is said to be equivariant to the action of a group $\mathscr{G}$ if $\mathscr{F}(\mathscr{T}_g(\mathscr{X})) = \mathscr{S}_g(\mathscr{F}(\mathscr{X}))$ for all $g \in \mathscr{G}$, where $\mathscr{T}_g, \mathscr{S}_g$ are transformations parameterized by g.

Invariance. A function $\mathscr{F}$ is said to be invariant to the action of a group $\mathscr{G}$ if $\mathscr{F}(\mathscr{T}_g(\mathscr{X})) = \mathscr{F}(\mathscr{X})$ for all $g \in \mathscr{G}$, where $\mathscr{T}_g$ is a transformation parameterized by g. Invariance is a special case of equivariance, where $\mathscr{S}_g$ is a trivial group action (i.e., identity).

In our setting, $\mathscr{F}$ is a neural network for processing an input molecule $\mathscr{X}$. The structure of the $\mathscr{F}$ defined on $\mathscr{X}$ is influenced by the symmetry of the underlying domain. This symmetry serves as a strong inductive bias, leading to more efficient learning by limiting the possible interpolants, $\mathscr{F}(\mathscr{X})$, to only those that adhere to the symmetry priors (Bronstein et al., 2021). For instance, molecular dynamics simulation aims to predict the future positions of the atoms given the current system state. Intuitively, the positions generated by a neural network should rotate, translate, and reflect together with the current system. Consequently, the inductive bias is employed by designing the neural network that transforms equivariantly under the actions of E(3). As another example, 3D molecular graphs are characterized by the absence of any inherent ordering of the nodes, and thus any computations performed on graphs should be agnostic to the ordering of nodes. To achieve this desirable property, neural networks operating on graphs must be permutation invariant, meaning their outcomes remain identical for any

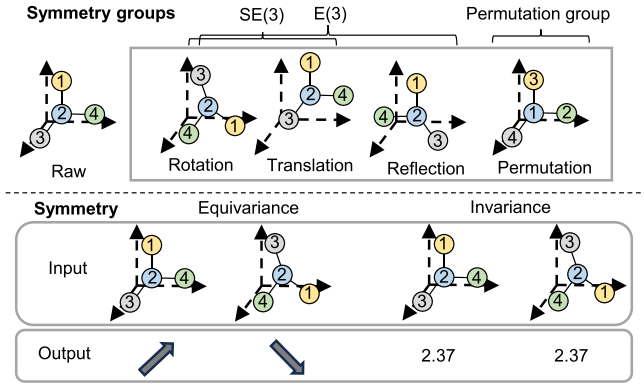

Fig. 3. Illustration for symmetry group and symmetry. Symmetry groups (upper portion), each encompass its own set of transformations. In the context of rotational symmetry(lower portion), it refers to the scenario where the vector outputs rotate along with the input molecule, while the scalar outputs remain unaffected.

two isomorphic graphs.

Symmetry has been extended to the probability distribution. Assuming that the probability distribution of variable x is denoted by $p(x)$. Invariance and equivariance of distribution are formulated as follows:

Invariant distribution. A distribution $p(x)$ is invariant to the action of a group $\mathscr{G}$ when $p(\mathscr{T}_g(x)) = p(x)$ for all $g \in \mathscr{G}$, where $\mathscr{T}_g$ is a transformation parameterized by g.

Equivariant distribution. A conditional distribution $p(y|x)$ is equivariant to the action of a group $\mathscr{G}$ when $p(\mathscr{T}_g(y)|\mathscr{T}_g(x)) = p(y|x)$ for all $g \in \mathscr{G}$, where $\mathscr{T}_g$ is a transformation parameterized by g.

A generative task aims to model $p(x)$, in some cases $p(x)$ are required to be unaffected by actions of a predefined symmetry group(e.g., the probability distribution of molecular conformation is unaffected by translation and rotation in molecular conformation prediction task). If $p(x)$ is invariant and $p(y|x)$ is equivariant, then the marginal distribution of y at any time step is also invariant to the group transformations, according to which we can build an equivariant generative model(Köhler, Klein, & Noé, 2020; Xu et al., 2022).

3. Molecular property prediction

In this section, we present the work of predicting molecular properties, such as quantum properties, physicochemical properties, and biological properties. In small molecular property prediction tasks, molecules are generally represented as atom-level point clouds or 3D graphs.

Invariant representations are used to predict quantum properties. For example, SchNet(Schütt et al., 2017) injects interatomic distances into the process of message passing, and the learned representations are E(3)-invariant as the result of interatomic distances have the property of E(3)-invariance. DimeNet(Gasteiger et al., 2020) combines distance and angle information with the message passing stage. Based on DimeNet, GemNet(Gasteiger et al., 2021) further incorporates dihedral angle information and involves 2-hop messages during the message passing stage, which makes the learned representations SE(3)-invariant. By constructing a local spherical coordinate system, SphereNet(Liu et al., 2022) realizes SE(3)-invariant message passing by using distance, angular, and torsional information, which is capable of recognizing the chirality property.

However, invariant message passing by utilizing only distances and bond angles results in an information loss. For instance, consider two atom arrangements that differ solely in their dihedral angles. The representations learned from invariant messaging using distances and bond angles would be identical for both arrangements. However, the representations should differ to fully capture the distinct dihedral angle configurations. This indicates that invariant messaging confined to distances and bond angles fails to encode all relevant information about local environments (Schütt et al., 2021). Thus, some methods adopt equivariant representations through equivariant message passing. EGNN(Satorras et al., 2021) proposes an E(n)-equivariant graph neural network, it obtains an equivariant message by multiplying the invariant message with an equivariant vector. GMN(Huang et al., 2022) extends EGNN with multiple equivariant vectors(e.g., positions, velocities, and forces) as inputs to model a more complicated system. TorchMD-NET (Thölke and Fabritiis, 2022), an O(3)-equivariant transformer architecture, leverages the same idea based on multiplication to conduct information exchange between invariant and equivariant features. Different from the methods mentioned above, some methods(Anderson, Hy, & Kondor, 2019; Batzner et al., 2021; Brandstetter, Hesselink, van der Pol, Bekkers, & Welling, 2022; Fuchs et al., 2020; Thomas et al., 2018) construct equivariant message passing combined with spherical harmonics and Wigner-D matrices, however, the computational overhead of these methods is generally expensive. There is a review that summarizes the approaches for implementing equivariant GNNs(Han et al., 2022). Commonly, equivariant GNNs perform better than those that only consider invariance on the task of predicting quantum properties (Miller et al., 2020; Brandstetter et al., 2022).

As for biological and physicochemical properties, SGCN(Danel et al., 2020) proposes graph convolutional networks(GCN) that exploit geometric features to predict physicochemical properties. GeomGCL(Li et al., 2021b) employs contrastive learning to predict biological and physicochemical properties, it proposes dual-channel message passing neural networks to capture distance and angle information under both 2D graphs and 3D graphs. Rather than representing molecules as 3D graphs, Drug3D-Net(Li et al., 2021a) first represents a molecule as 3D voxels by setting each channel to a different atom type. Based on 3D CNNs, it proposes a spatial–temporal attention module, which extracts geometric features in the process of convolution for predicting the biological and physicochemical properties of molecules.

4. Intermolecular interaction

Within the realm of intermolecular interaction, we are engaged in two subtasks: identifying protein binding sites and predicting protein binding affinity.

4.1. Binding site prediction

Protein binding site prediction is a meaningful task in vivo. In GDL, binding sites of protein structure can be represented in various forms — residue level nodes/points, voxels, and some areas of the protein surface.

ScanNet(Tubiana et al., 2021) represents proteins as point clouds of atoms and amino acids to predict protein–protein binding sites, atom-level and amino-acid-level features are extracted with spatial-chemical filters in local coordinate frames which guarantee E(3)-invariance. Geometric Transformer (Morehead et al., 2022) incorporates SE(3)-invariant graph transformer to predict protein interface contacts. Given a pair of residue-level protein graphs, Geometric Transformer is trained to evolve geometric features of protein structures simultaneously with coevolutionary features to predict inter-chain interface contacts. PointSite(Yan et al., 2022) tackles protein–ligand binding sites prediction as the task of point cloud segmentation, representing the protein–ligand complex as an atom-level point cloud and identifying the atoms that belong to ligand binding atoms, Those atoms are used to predict protein–ligand binding sites.

Given a fixed-voxel grid labeled as positive if its geometric center is closer than a predefined threshold to the binding pocket geometric center and negative otherwise, DeepSite(Jiménez et al., 2017) outputs the binding site label probability with a 3D convolution neural network.

Also based on 3D CNNs, SAPocket(Wang et al., 2023) first samples points from the protein surface and voxelizes each point along with its surrounding environment. Then, these voxelized points are scored using 3D CNNs. Finally, the sample points with high scores are clustered to predict protein pockets.

MaSIF(Gainza et al., 2019) decomposes a protein surface into patches with fixed geodesic radius, wherein each vertex is assigned with geometric and chemical features, geodesic convolutions are used to extract patch-level descriptors which can be processed to predict protein–protein interaction sites. dMaSIF(Sverrisson et al., 2021), an end-to-end differential approach, accelerates such pattern by directly operating on points on the surface without considering connections and expensive precomputation, quasi-geodesic convolutions are then used to extract features from such point clouds, SE(3)-invariance of the model is achieved by only considering local chemical and geometric properties of the molecular surface.

4.2. Binding affinity prediction

The protein–ligand binding affinity is used to measure the strength of a ligand binding to its target protein. Measuring the interaction between proteins and ligands is the key to binding affinity prediction. Generally, there are two common methods to represent protein–ligand complex — voxel-based and graph-based methods.

Voxel-based methods represent protein–ligand complex as 3D voxels by cropping the region of interest with a bounding box and encoding information of the complex into voxels, 3D convolution neural networks are used to extract features from the voxels. AtomNet(Wallach et al., 2015) predicts the bioactivity of small ligands by injecting the information of atom types and protein–ligand complex descriptors into 3D voxels. K_{DEEP}(Jiménez et al., 2018) predicts binding affinity by encoding physicochemical and structural features into voxels.

The protein–ligand complex can be naturally regarded as a graph where nodes correspond to atoms/residues and edges correspond to bond/non-bond interactions. Torng et al.(Torng and Altman, 2019) proposed an approach that utilizes an unsupervised deep graph auto-encoder to learn protein pocket representation, and graph convolution networks are used to extract fingerprints of protein pockets and ligands respectively, these fingerprints are then concatenated to predict the activity of the protein–ligand pair.

A general graph-based pattern to learn from the protein–ligand complex is that(Fig. 4), for a given 3D graph of a protein–ligand complex, covalent interactions usually exist inside the molecule, while non-covalent interactions exist between molecules which are generally determined by Euclidean distances, such intramolecular and intermolecular interactions are captured by taking message passing respectively, the learned node or edge features are used for the target tasks. For example, by incorporating covalent messages in the phase of intramolecular interactions and Euclidean distances in the phase of intermolecular interactions, PIGNet(Moon et al., 2020) and InteractionGraphNet(Jiang et al., 2021) sequentially learn intramolecular interactions and intermolecular interactions to predict binding affinity, Lim et al.(Lim et al., 2019) predicted whether a ligand is active against the target by using graph attention neural networks to capture intermolecular and intramolecular interactions separately. GIGN(Yang et al., 2023) proposes a SE(3)-invariant heterogeneous layer that processes intermolecular and intramolecular interactions independently during message passing to predict binding affinity.

HOLOPROT(Somnath et al., 2022) is a method used for predicting protein–ligand binding affinity and protein properties. It achieves this by learning a multi-scale graph representation from various sources such as the protein surface, sequence, and structure by allowing the high-level representation to integrate information from the low-level representation. Specifically, the protein surface is represented as a mesh, while the primary, secondary, and tertiary structures of the protein are represented as a graph where two C_α atoms are connected if the Euclidean distance between them is lower than a predefined threshold, multi-scale message passing networks. are then used to extract features from them. SIGN(Li et al., 2021c) represents the protein–ligand complex as a 3D graph taking interatomic distance as edges to predict binding affinity. To this end, it proposes a polar-inspired graph attention mechanism that allows for message passing in polar coordinates. Furthermore, the method incorporates self-supervised training over interactive edges to capture global interactions.

4.3. Molecular design

Molecular design tasks can generally be divided into three categories: ligand-based drug design(LBDD), structure-based drug design (SBDD), and macromolecule design(or protein). Both SBDD and LBDD are computational approaches utilized to rationally design novel small molecule ligands targeted against specific biomolecules of interest. When the 3D structure of a protein implicated in a disease is available, SBDD is commonly utilized to develop targeted therapeutics. SBDD leverages the 3D shape and architecture of the protein target as a template for rationally designing novel drug candidates. However, SBDD is contingent on solved protein structures, which are unavailable for some discovery projects. In such cases, LBDD offers an alternative approach, wherein known binders of the target are analyzed to derive insights guiding the design of enhanced ligands with potential as therapies. Thus, SBDD and LBDD represent complementary techniques in drug development pipelines, with the former relying on target structure and the latter relying on known active ligands in the absence of structural data.

4.4. Ligand-based drug design

LBDD denotes the process of identifying potential therapeutic compounds in the absence of target protein structures while relying on the chemical structures of known molecules that can modulate the target.

Most recent methods generate molecules in an autoregressive pattern. For instance, G-SchNet(Gebauer et al., 2019), an autoregressive model, generates 3d point sets from scratch for the targeted discovery of

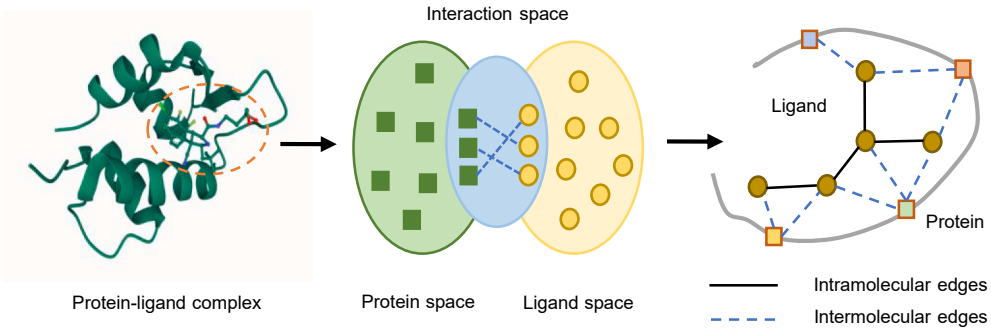

Fig. 4. A common pattern of modeling protein–ligand interactions. Interactions within and between protein and ligand are considered separately.

molecules while ensuring SE(3)-invariance of atom type distribution and SO(3)-equivariance of atom coordinate distribution. Based on G-SchNet, 3D-Scaffold(Joshi et al., 2021) discovers novel drug candidates with desired properties by directly taking the corresponding desired functional group as input and then optimizing it. Building on G-SchNet, cG-SchNet(Gebauer et al., 2021) generates molecules with specified chemical and structural properties and permits designing molecules with any desired property values after training is done. Rather than directly generate atom coordinates, G-SphereNet(Luo and Ji, 2022) first generates 3D molecular geometries by predicting distances, angles, and torsion angles which satisfy SE(3)-invariance, consequently, atom coordinates are determined and satisfy SE(3)-equivariance. Some works (Simm, Pinsler, Csányi, & Hernández-Lobato, 2021; Simm, Pinsler, & Hernández-Lobato, 2020) adopts deep reinforcement learning to design molecules by injecting fundamental physical properties like energy to reward function, They sequentially place atom in internal coordinates which is relative to previously placed atom to achieve covariance of rotation and translation.

E-NF(Garcia Satorras et al., 2021) first extends the flow-based model to generate molecular point clouds in 3D space, specifically, E-NF models an invertible equivariant flow by formulating it as a solution to the ordinary differential equation. Denoising Diffusion Probabilistic Models (DDPM)(Ho et al., 2020; Sohl-Dickstein et al., 2015), a class of generative models, shine brightly in the field of Drug Design. EDM (Hoogeboom et al., 2022) extend the diffusion model to generate molecules in 3D space by proposing an E(3)-equivariant diffusion model directly operating on atom coordinates and categorical features.

4.5. Structure-based drug design

SBDD aims to design novel molecules that bind to a target protein with high affinity and specific(A. C. Anderson, 2003). Most of the recent works represent molecules as 3D graphs. By combining a rotational equivariant network with Monte Carlo Tree Search, DeepLigBuilder(Li et al., 2021d) samples atoms sequentially to construct a new molecule. Similarly, 3DSBDD(Luo, 2022) directly uses SE(3)-invariant GNN to predict the new atom type sequentially, GraphBP(Liu et al., 2022) generates atom type and relative location w.r.t local coordinates system to preserve equivariance property. Instead of sampling atoms, Powers et al. propose an E(3)-equivariant neural network to sample fragments and dihedral angles determining attachment geometry(Powers et al., 2022). Inspired by the diffusion model, DiffSBDD(Schneuing et al., 2022) and TargetDiff(Guan et al., 2023) propose an equivariant diffusion model to generate small ligands within a protein pocket.

Different from the above work, LiGAN(Ragoza et al., 2021) represents the binding site and molecule as 3D atom density, introducing a conditional variational auto-encoder(VAE)(Kingma and Welling, 2013; Pagnoni et al., 2018) model to generate molecules condition on binding sites by extracting features with 3D CNNs. GEOM-CVAE(Li et al., 2022) propose a constrained VAE for molecular generation by converting 3D coordinates of drug molecules into images and modeling the protein surface as a mesh.

Linker Design, a related task, means designing a small linker attaching several independent molecules. 3DLinker(Huang et al., 2022) generate linker for a pair molecules with an E(3)-equivariant graph variational autoencoder framework. DiffLinker(Igashov et al., 2022) generates a linker for a set of disconnect fragments while taking the surrounding pocket into account by an E(3)-equivariant diffusion model.

4.6. Macromolecular design

Generally, protein design tasks associated with protein structure can be categorized into three main methodologies. The first approach involves the direct design of the protein backbone. The second approach entails generating sequences that, based on the protein's structure, can be folded into this structure (a.k.a inverse folding). The third approach involves the concurrent design of both the protein structure and sequence.

In the case of directly generating protein structure, Anand et al. (Anand and Huang, 2018) take advantage of Generative Adversarial Networks(GAN) to generate pairwise distances between α- carbons on the protein backbone, they additionally present a method that SE(3)-invariantly determines 3D cartesian coordinates according to generated pairwise distances. Ig-VAE(Eguchi et al., 2020) employs VAE to construct class-specific backbones by using VAE to directly generate 3D coordinates SE(3)-invariantly. ProtDIFF(Trippe et al., 2023) designs an E(3)-equivariant diffusion model that can generate realistic protein backbones, it further proposes a sampling algorithm to generate scaffolds conditioned on the given motif with the diffusion model.

In the case of inverse folding, GVP-GNN(Jing et al., 2021) represents protein as an amino acid level graph where each node is connected to its K nearest neighbors based on the distance between their C_α atoms, it introduces geometric vector perceptrons that can operate both on scalar and geometric features in an invariant and equivariant manner respectively to learn node representations, which are used to autoregressively generate the amino acid sequence. Structured Transformer(Ingraham et al., 2019) proposes a transformer-based encoder to obtain SE(3)-invariant features of protein structures and autoregressively decodes them into amino acid sequences.

In the case of co-designing the 3D structure and sequence, RefineGNN(Jin et al., 2022) co-designs the 3D structure and sequence of the antibody. Specifically, it autoregressively generates protein sequences and refines its predicted protein structure after each residue type is determined, and the refined protein structure guides the subsequent residue type choices. MEAN(Kong et al., 2023) generates the 1D sequence and 3D structure of an antibody conditioned on its target antigen and the light chain of the antibody, in which an attentive E(3)-equivariant message passing neural network is used to capture geometric connection in and between these components. Inspired by AlphaFold2(Jumper et al., 2021), PROTSEED(Shi et al., 2023) proposes a trigonometry-aware encoder that infers geometric constraints and interaction from single and pair context features. A SE(3)-equivariant decoder is then used to translate protein structure and sequence independently.

4.7. Molecular conformation prediction

Molecular conformation prediction aims to predict the 3D structure of molecules. According to the differences in the generated objects, we divide the molecular conformation prediction task into three categories: molecule-based conformation prediction (for small molecules), complex-based conformation prediction (for multi-molecular complex), and protein-based conformation prediction (for proteins).

4.8. Molecule-based conformation prediction

The task of molecule-based conformation prediction is a conditional generation problem. Given a 2D graph $\mathscr{G}$, molecule-based conformation prediction means to predict 3D coordinates of atoms of molecule $\mathscr{R}$, which can be represented as $p(\mathscr{R}|\mathscr{G})$.

As an early attempt to predict molecular conformation with deep learning, CVGAE(Mansimov et al., 2019) uses conditional VAE to directly predict atom coordinates, but it fails to respect essential symmetry (e.g. SE(3)-equivariance) and doesn't perform well. Instead of generating atomic coordinates directly, some molecular conformation prediction methods produce intermediate geometric features. These features include interatomic distances or torsion angles, which are then used to predict the atomic coordinates. GraphDG(Simm and Hernández-Lobato, 2020) and CGCF(Xu et al., 2021) respectively take advantage of the VAE model and Flow-based model to predict a set of interatomic distances, then convert them to a set of atomic coordinates with classic Euclidean Distance Geometry algorithm. ConfVAE(Xu et al., 2021)

improves upon a two-step pattern used for conformation prediction by utilizing bilevel optimization to enable end-to-end training of the model.

GeoMol(Ganea et al., 2021) predicts local molecular structure by predicting bond distances and bond angles while maintaining permutation equivariance with respect to neighboring atoms. To ensure SE(3)-invariance, the model then predicts SE(3)-invariant torsion angles to assemble the local structures. By doing this, the model avoids the unnecessary over-parameterization of the geometric degrees of freedom.

ConfGF(Shi et al., 2021), DGSM(Luo et al., 2021) learn SE(3)-equivariant gradients field of atomic cartesian coordinates via a score matching method only relying on interatomic distance, DGSM additionally models long-range interactions between non-bond atoms by dynamically constructing graph based on spatial proximity.

GeoDiff(Xu et al., 2022) extends the diffusion model to directly predict atom coordinates of small molecules, an SE(3)-equivariant reverse generation process is carefully designed to respect the symmetry of the system. DMCG(Zhu et al., 2022) generates molecular conformations with VAE, The generated conformations are refined by following the backbone model that applies a loss function that is invariant to the transformation of roto-translation and permutation of symmetric atoms.

4.9. Complex-based conformation prediction

Complex-based conformation prediction means to predict the complex structure of a small ligand or a protein binding to the target protein (a.k.a docking).

DeepDock(Méndez-Lucio et al., 2021) represents the protein surface as a polygon mesh and the ligand as a graph to predict the distribution of the distance between the protein node and the ligand node by extracting features with graph convolutions.

By representing proteins as residue-level 3D graphs and ligands as atom-level 3D graphs, some methods predict binding conformation via message passing neural networks. EquiDock(Ganea et al., 2022) executes protein–protein docking by predicting key points for each protein represented by an atom-level 3D graph and alignment them, in which they propose Independent E(3)-Equivariant Graph Matching Network (IEGMN) utilizing inter- and intra-node messages to SE(3)-equivariantly update coordinates and SE(3)-invariantly update node features for a given pair of 3D graphs, EquiBind(Stärk et al., 2022) takes advantage of IEGMN to dock small flexible ligand with protein, additionally changes the torsion angle of rotatable angles from the initial conformer to match the generated conformer while keeping the plausibility. TankBind(Lu et al., 2022) explicitly addresses all potential binding sites for each protein and incorporates trigonometric constraints into the message passing process. It finally infers an intermolecular distance map, which is used for binding structure prediction. DiffDock(Corso et al., 2023) proposes a diffusion generative model to dock small ligand into a protein pocket, it develops a diffusion process on the product space of freedom (translational, rotational, and torsional) which considers the flexibility of the ligand pose.

HMR(Wang et al., 2023) represents molecular surface as a 2D Riemannian manifold and proposes a heat-diffusion-inspired spectral message passing over the manifold. This approach effectively extracts geometric and chemical features for binding site prediction and rigid protein docking.

Besides, some methods have been proposed to assess the generated docking models, DeepRank(Renaud et al., 2021) maps atomic and residue features on 3D voxels using the Gaussian function, and extracts features with 3D CNNs to classify protein–protein interfaces and assess protein–protein docking models.

4.10. Protein-based conformation prediction

The challenge of predicting protein structures at an experimental level of accuracy from amino acid sequences has been largely addressed by AlphaFold2(Jumper et al., 2021), as reported in the 14th Critical Assessment of Protein Structure Prediction (CASP 14). RoseTTAFold (Baek et al., 2021), ESMFold(Lin et al., 2022), and OmegaFold(Wu et al., 2022) are protein structure prediction methods that have emerged following the success of AlphaFold2. These methods leverage deep learning and large-scale protein sequence data to predict protein structures with high accuracy. While not as well-known as AlphaFold2, they still contribute to the advancement of computational protein structure prediction. EIGENFOLD(Jing et al., 2023), a recent work utilizing a diffusion generative model, models protein structure as a system of harmonic oscillators and predicts forces that are combined with harmonic constraints to update the residue positions iteratively during the denoising step.

In addition to the above conformation prediction work, abundant meaningful works have been proposed to analyze and understand the structure of large biomolecules. Proteins have a hierarchical structure: the primary structure is determined by amino acid sequence, the secondary structure arises from hydrogen bonds between distant amino acids, the tertiary structure reflects protein folding and the quaternary structure is the result of interactions between multiple amino acid chains. IEConv(Hermosilla et al., 2021) deals with fold classification and enzyme-catalyzed reaction classification, it considers structural levels of proteins (primary, secondary, tertiary) by using a novel convolution operator that combines Euclidean and geodesic distances on a multi-graph representation of proteins and designs a set of hierarchical pooling operators that enable multi-scale analysis. SpookyNet(Unke et al., 2021) uses an E(3)-equivariant message passing neural network to predict force fields of protein structures, explicitly modeling electronic degrees of freedom and nonlocality to consider chemically meaningful inductive biases. Besides, the accuracy of the calculated protein structure models is generally unstable, therefore some methods have been proposed to assess the generated models and select the model with high accuracy as the output result. Similar to protein structure, RNA structure can adopt complex three-dimensional structures that are essential for their biological functions. GVP-GNN(Jing et al., 2021) leverages the representations of all nodes to predict the quality of the protein model. ARES(Townshend et al., 2021) proposes a scoring function to assess the accuracy of structural models of RNA. Given the input of 3D coordinates and the chemical element type of each atom, each layer of ARES is SE(3)-equivariant and preserving the orientation and position of identified motifs, which can be used to recognize coarser-scale motifs.

4.11. Molecular 3D pretraining

In the field of drug discovery and life science, the chemical space of molecules is vast, while labeled data is limited. To obtain better molecular representation, one solution as shown in Fig. 5**a**, we can learn good representations from large-scale unlabeled molecules via pre-training(or self-supervised learning)(Bengio, Courville, & Vincent, 2012; Hamilton, Ying, & Leskovec, 2017; Zhang, Yin, Zhu, & Zhang, 2017), then we finetune the pre-trained model on limited labeled molecules to adapt specific tasks. Molecular 3D self-supervised learning can be generally achieved in four ways: predictive pretraining, generative pretraining, contrastive learning, and denoising (Fig. 5**b**).

Predictive pretraining aims to let the model predict self-supervised labels that can be acquired by experiments or precalculation. Such as molecular conformation and momentum (HamNet(Li et al., 2021e) and LagNet(Li et al., 2023)), force(Jiao et al., 2022), and energy(Luo et al., 2023). Generative pretraining aims to let the model reconstruct specified parts of the input molecule. For instance, geometric features(e.g. interatomic distance, angle, etc.)(Fang et al., 2021; Zhang et al., 2023), atom type(Uni-Mol(Zhou et al., 2023)), residue type(Zhang et al., 2023), molecular dynamic trajectory(PROTMD(Wu et al., 2022)), whole molecule according to its 3D shape (DESERT(Long et al., 2022)).

Contrastive learning considers multi-view or multi-modal input. The magic of contrastive learning lies in the training objective, which creates

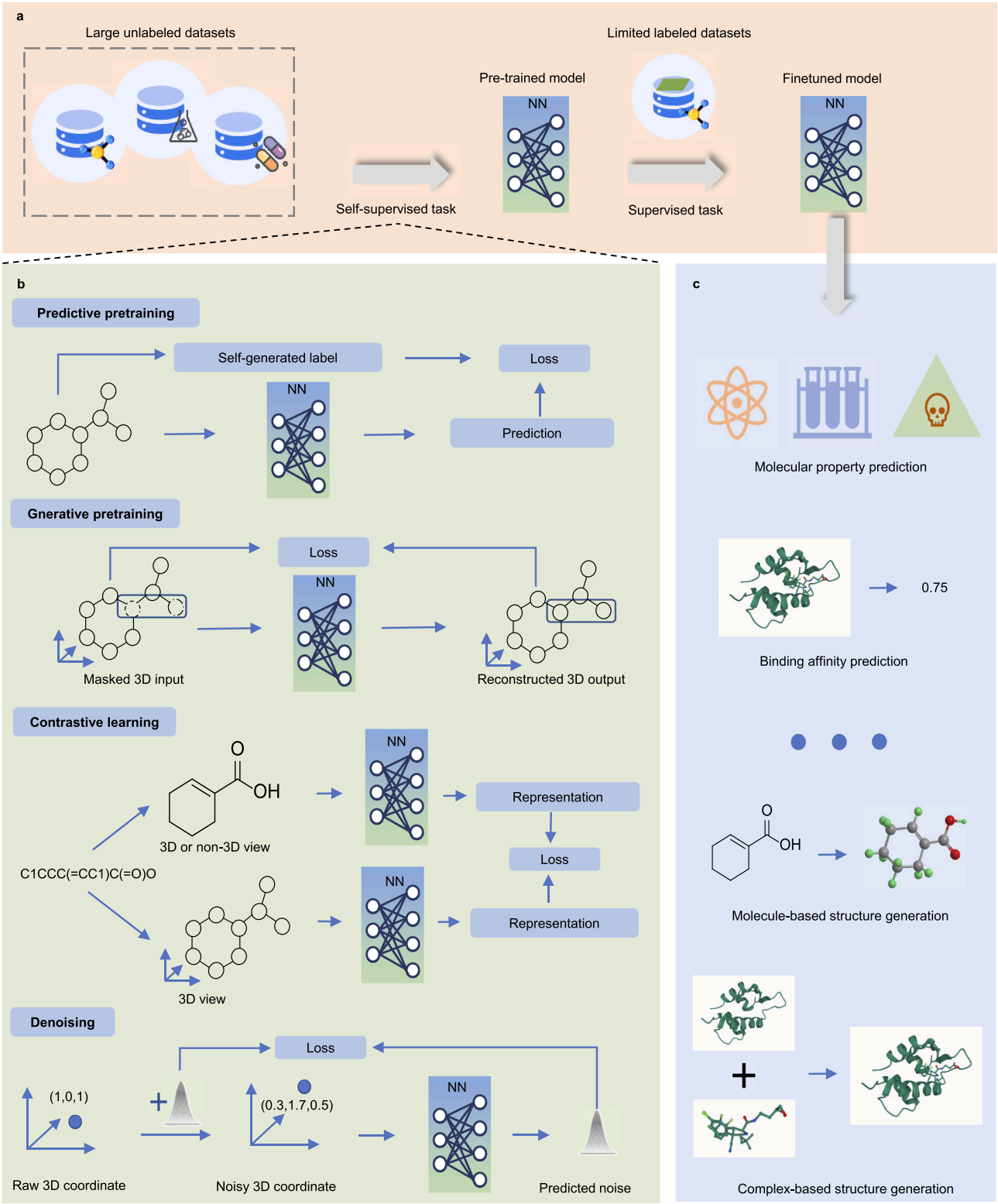

Fig. 5. Molecular 3D pretraining. (a) Overview of molecular 3D pretraining. **(b)** Self-supervised learning methods. **(c)** Downstream tasks.

a joint representation where paired molecular representations are pushed together and unpaired representations are pulled apart. For example, methods that take a 3D molecular graph and a 2D molecular graph(GrpahMVP(Liu et al., 2022),and (Stärk et al., 2022)), two 3D graph views of protein with different sampling scheme (GEARNET (Zhang et al., 2023)) as input.

Denoising in 3D space is a self-supervised learning method that adds random noise to the atomic coordinates of molecules and then trains the model to predict the noise. Denoising in 3D space has unique significance in the field of molecular pretraining. Assuming that the raw molecule is at the local energy minimum, imposing local and random noise on the geometry of this molecule leads to higher energy configuration, and predicting this noise is equivalent to learning a mapping that maps high energy to low energy(Godwin et al., 2021). Denoising tries to

learn a force field that plays an important role in corresponding geometry representation learning(Liu, Guo, & Tang, 2023; Zaidi et al., 2023).

After the process of leveraging self-supervised learning methodologies, a pre-trained model is generated. This model is subsequently fine-tuned for a range of downstream tasks (Fig. 5**c**). Such as physicochemical and biological property prediction(Fang et al., 2021; Li, Yang, Song, & Cai, 2021e; Liu et al., 2022; Zhou et al., 2023), quantum property predictions(Li, Yang, Song, & Cai, 2021e; Liu et al., 2022; Liu, Guo, & Tang, 2023; Zaidi et al., 2023; Zhou et al., 2023), LBDD(Long et al., 2022), binding affinity prediction(Liu et al., 2022; Wu et al., 2022), protein function prediction and fold classification(Zhang et al., 2023), molecule-based conformation prediction and complex-based conformation prediction(Zhou et al., 2023).

4.12. Outlook

Molecular descriptors. The evolution of molecular descriptors has been crucial for advancing the understanding and modeling of macromolecules. Despite their potential, 3D voxels have not garnered significant attention in recent years, primarily due to their computational complexity and lack of properties such as SE(3)-equivariance. Alternatively, 3D surfaces have been employed to represent molecular surfaces, warranting further investigation and development. Integration of computer graphics knowledge could enhance the expressiveness of molecular surface descriptors. While 3D voxels and 3D surfaces are predominantly utilized for macromolecular modeling, 3D graphs offer a versatile approach suitable for both small and large molecules. Currently, and for the foreseeable future, 3D graphs are expected to remain the predominant method for molecular 3D representation.

Challenges of tasks. The architecture based on irreducible representations and tensor products proves to be computationally intensive, making the reduction of training time and increased training efficiency a vital research direction. In the task of binding conformation prediction, most existing methods overlook the flexibility of ligands and protein pockets or only consider ligand flexibility. A more comprehensive approach would involve accounting for the flexibility of both receptors and ligands, potentially incorporating molecular dynamics simulation knowledge. To foster trust and encourage the adoption of GDL models in molecular and protein applications, enhancing interpretability is essential. Researchers are currently developing techniques to gain insights into the learned representations, such as determining key features that contribute to the model's predictions.

Equivariant generative model paradigm. In the context of generative modeling, the equivariant generative model paradigm plays a crucial role in generation tasks. Accounting for the equivariance of a system serves as a significant bias and enables models to adapt effectively to various transformations in the input data. Consequently, the adoption of equivariant generative models is likely to become a prevailing trend in future advancements within the field of generative modeling.

Increasing focus on macromolecules. The advent of AlphaFold2 has facilitated access to more accurate 3D structural information, which can be harnessed to train models and achieve enhanced performance. Examples of applications that stand to benefit from these advancements include 3D protein pretraining models, de novo protein design, and antibody design.

5. Terminology box

Geodesic Convolution(Masci et al., 2015): A generalization of the convolutional network paradigm to non-Euclidean manifolds.

Transformer(Vaswani et al., 2017): A deep learning model that uses a self-attention mechanism that can assign different weights according to the importance of each part of the input data.

Flow-based Model(Papamakarios et al., 2019): A generative model that creates new data by using normalizing flow. This method transforms a simple probability distribution into a more complex one using the change-of-variable law of probabilities.

Variational Auto-Encoder(VAE)(Kingma and Welling, 2013): A generative model that can learn latent attributes and construct new elements from the probability distribution of the hidden variable space.

Diffusion Model(Ho et al., 2020)**:** A generative model inspired by non-equilibrium thermodynamics. By learning the process of reverse diffusion, it can generate desired data samples by starting from noise.

Denoising pre-training(Zaidi et al., 2023): A pre-training method that adds random noise to input data and trains the model to predict the noise.

Score Matching(Song and Ermon, 2019): A statistical technique used in machine learning and probability theory to estimate the parameters of a probability distribution. It involves finding the model parameters that minimize the difference between the gradients of the model's log-density function and the log-density function of the data.

Optimal Transport: A mathematical framework used to quantify the minimum cost of transforming one distribution of items into another.

Euclidean Distance Geometry(Liberti et al., 2012): Euclidean distance geometry involves exploring Euclidean geometry based on the concept of distance.

Bilevel Optimization(Colson et al., 2007): Bilevel Optimization includes two levels of optimization tasks, one nested within the other. The optimization problem of the outer layer is usually referred to as the upper level optimization problem, while the optimization problem of the inner layer is usually referred to as the lower level optimization problem.

CRediT authorship contribution statement

Mingquan Liu: Data curation, Writing – original draft, Writing – review & editing, Visualization. **Chunyan Li:** Writing – review & editing, Visualization, Supervision. **Ruizhe Chen:** Data curation. **Dongsheng Cao:** Conceptualization. **Xiangxiang Zeng:** Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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